Annamneedi Suresh Kumar

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SKILLS

- **Machine Learning & Deep Learning:** PyTorch, Scikit-Learn, Supervised/Unsupervised Learning, Neural Networks, Model Evaluation
- **Data Preprocessing & Analytics:** Pandas, NumPy, Feature Engineering, Exploratory Data Analysis (EDA), Statistical Analysis
- **Generative AI & Vector Search:** LLM APIs (Gemini, OpenAI, Groq), RAG Retrieval, ChromaDB, FAISS Vector Search, CLIP Embeddings
- **Programming Languages:** Python (Clean & Modular), JavaScript (ES6+), C++, SQL (PostgreSQL), HTML5/CSS3
- **Backend & API Integration:** FastAPI, Flask, RESTful APIs, Node.js, Express.js, Docker, Cloud Platforms (OCI Certified)
- **CS Fundamentals:** Data Structures & Algorithms, Object-Oriented Programming (OOP), Git Version Control, Operating Systems

PROJECTS

IntelliMoE — Multi-Agent AI Orchestration & Model Evaluation Platform | GitHub — Jun 2026

- Architected an autonomous Mixture-of-Experts (MoE) orchestration platform in Python, integrating Gemini, Groq, and OpenAI LLMs with hybrid semantic query routing, RAG retrieval pipelines, and ChromaDB vector embeddings.
- Engineered a comprehensive model evaluation system benchmarking latency, throughput, token costs, and factual response accuracy across multi-provider endpoints; implemented validation guardrails to prevent hallucinated outputs.
- Conducted error analysis across agentic task outputs to improve response reliability; containerized the execution environment using Docker Compose with automated Pytest test suites.
- **Tech:** Python, Generative AI, Mixture-of-Experts (MoE), LLM APIs, RAG, ChromaDB, Model Evaluation, Docker, Pytest, REST APIs.

Edge Deep Learning Model Compression — Optimization Benchmark | GitHub — Sep 2026

- Engineered a PyTorch benchmarking framework evaluating machine learning optimization techniques (INT8 post-training quantization, structured pruning, knowledge distillation) on ResNet-18 deep learning architectures.
- Designed systematic data preprocessing and ablation pipelines using NumPy and Pandas to analyze validation loss, parameter pruning ratios, and CPU inference latency across varying computational constraints.
- Achieved an 87.36% model storage reduction (85.70 MB to 10.83 MB) via INT8 quantization with only 0.36% accuracy loss while accelerating inference speed by 1.85x–2.5x (latency to 8.38 ms/image); served via FastAPI endpoints.
- **Tech:** Python, PyTorch, Model Optimization, Quantization, Pruning, Distillation, Data Science, Benchmarking, FastAPI.

TailorTalk — Multimodal AI Reasoning & Vector Retrieval Engine | GitHub — Jan 2026

- Developed a multimodal conversational agent in Python leveraging OpenAI CLIP (ViT-B/32) to generate normalized 512-dimensional vision-language embeddings for zero-shot semantic querying and feature extraction.
- Built a high-performance vector retrieval pipeline using FAISS (IndexFlatIP) combined with custom color histogram reranking heuristics and a LangChain conversational agent loop for interactive user queries.
- Implemented contextual memory management to preserve conversational state across multi-turn interactions, achieving sub-15ms vector retrieval latencies across high-dimensional feature spaces.
- **Tech:** Python, Multimodal AI, LangChain, OpenAI CLIP, FAISS Vector Search, Feature Engineering, Contextual Memory, REST APIs.

StockFlow — Full-Stack Data Pipelines & Observability Infrastructure | GitHub — Feb 2026

- Architected an enterprise transactional inventory platform utilizing React 18, TypeScript, and FastAPI, implementing real-time state mutation tracking, automated inventory alerts, and transactional data integrity.
- Engineered relational PostgreSQL schemas with composite indexing, transaction isolation, and JWT role-based access control; established automated multi-branch CI/CD workflows using GitHub Actions and Docker Compose.
- Operationalized real-time system observability using Prometheus metric scraping and Grafana dashboards, tracking API request latencies, database query throughput, and system health under simulated loads.
- **Tech:** Python, FastAPI, TypeScript, PostgreSQL, Relational Data Modeling, Docker, Nginx, Prometheus, Grafana, GitHub Actions.

ACHIEVEMENTS & RESEARCH

- **Code Off Duty Hackathon** — Qualified for Round 2 of the Code Off Duty Web Hackathon organized by Coding Wise, building and defending full-stack prototypes and algorithmic solutions under competitive time constraints. | Mar 2025
- **Data Structures and Algorithms Using C++** — Completed 3-month comprehensive training (Jun 2025 – Aug 2025) mastering graph theory, dynamic programming, priority queues, and asymptotic complexity optimization for robust software design.

CERTIFICATES

- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate | Oracle University — Mar 2026
- Principles of Generative AI | Infosys Springboard — Dec 2025
- Master Generative AI & Tools | Udemy — Aug 2025

EDUCATION

- ***Lovely Professional University*** — **Punjab, India**
  Bachelor of Technology, Computer Science and Engineering (Artificial Intelligence & Machine Learning); CGPA: 8.32 — Aug 2023 – Jul 2027
- ***Sri Shiridi Sai Jr College*** — **Rajahmundry, Andhra Pradesh**
  Intermediate (MPC); Percentage: 87.8% — Jun 2021 – Apr 2023
- ***Vignan School*** — **Malikipuram, Andhra Pradesh**
  Secondary School; Percentage: 81% — Jun 2019 – Apr 2021